Siddharth Korukonda

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Education

Stony Brook University 2023 – 2027
Bachelor of Science in Computer Science, Bachelor of Arts in Economics (Dean's List) *Stony Brook, NY*
Minor in Applied Mathematics and Statistics, Specialization track in Artificial Intelligence and Machine Learning

Courses: Theory of Computation, Analysis of Algorithms, Data Structures, Object-Oriented Programming, Systems Fundamentals, Programming Abstractions, Linear Algebra, Calculus I–IV, Probability Theory, Data Mining, Game Theory, Graph Theory, Machine Learning, Artificial Intelligence, Logic, Scripting

Awards and Certifications

Mathematical Association of America (MAA): AIME - 5 (AMC 12B - 106.5)
Hackathons Won: Columbia DivHacks 2025, SBUHacks 2025, HopperHacks 2026
JP Morgan Chase: Quantitative Reserach Job Simulation, Software Engineer Job Simulation
Amazon: Certified Cloud Practitioner (in progress)
Google: Data Analytics Professional (in progress)

Technical Skills

Languages: Python, Java, Kotlin, MySQL, C/C++, C#, Lua, JavaScript, HTML/CSS, Bash, TypeScript, LaTeX, R, MIPS
Frameworks: React.js, Node.js, Express.js, .NET, Django, Spring Boot, MongoDB, PyTorch, Flask, Tailwind, REST APIs
Developer Tools: Postman, Git, LangChain, Hugging Face, PyCharm, IntelliJ, Kubernetes, Docker, AWS CLI, Jira, Pinecone, Snowflake
Libraries: NumPy, Matplotlib, Scikit, SciPy, Pydantic, OpenCV, TensorFlow, OpenPose, Asyncio, ONNX Runtime, Transformers, Pandas
Concepts: NLP, CI/CD, Computer Vision, Software Security, Operating Systems, Cloud Computing, AWS, MCP Servers, Time Series Analysis, Regression Analysis, Statistic Analysis, Market Modeling

Experience

Undergraduate Teaching Assistant January 2026 - Present
Stony Brook University *Stony Brook, NY*
- Created assignments and labs for CSE 220 (System Fundamentals I).
- Delivered assistance to students in lectures of approximately 200 students.
- Graded assignments and held weekly recitations and office hours.

Software Developer January 2025 – Present
Stony Brook University Hospital *Stony Brook, NY*
- Design and build the hospital website in React and Flask, create reusable UI components, develop secure APIs, manage SQL and NoSQL data, implement login and roles, and keep pages fast and accessible.
- Collaborate with BS/MD and pre-med students to analyze de-identified patient records, curate datasets, and train/evaluate clinical AI models, incorporating feedback to improve model quality.
- Write clear docs, help teammates and staff use the tools, watch error logs, and fix issues to keep everything stable.

Secure Distributed Computation and Learning Networks Research August 2024 – Present
Undergraduate Researcher *Stony Brook, NY*
- Maintained 80% consensus in Byzantine networks by projecting updates onto convex sets and eliminating adversarial values
- Achieved 77% valid input preservation in real-time decision-making under adversarial corruption
- Enabled stable updates with 93% agent interaction through controlled neighborhood protocols

Research Papers

NFLPA Analytics – Special Teams Workload and Injury Risk 2025 - 2026
Python, DuckDB, Statsmodels, Poisson and Negative Binomial GLMs, Fixed Effects
- Analyzed special teams workload and next-week injury risk using **5,950 NFL team-weeks**, estimating fixed-effects Poisson and Negative Binomial models with clustered and bootstrap-validated inference.
- Shock workload weeks raised next-week injury probability by **2.52 pp (offense)** and **1.39 pp (defense)**, while workload volatility (25th–75th percentile) increased offensive injury risk by **2.46 pp** with no meaningful defensive effect.
- League-level scaling implies **+8.8 offensive** and **+5.7 defensive injuries per season** attributable to shock workloads; results documented in a research paper submitted to the NFL Players Association.

Projects

SurveiLens - Real-Time AI Safety Platform (Hackathon Winner) November 2025
React, Flask, NeuralSeek, Snowflake, ElevenLabs, OpenCV, Gemini, VADER, WebRTC/RTSP, Vultr
- Designed and implemented a real-time agentic and multimodal AI system that fuses video, audio, and language signals to generate continuous risk assessments from live streams.
- Engineered inference pipelines using OpenCV, Gemini-based reasoning, and sentiment analysis to improve detection accuracy and reduce false positives across modalities.
- Deployed scalable RTSP/WebRTC ingestion with Snowflake-backed event storage and NeuralSeek-orchestrated alerting to support low-latency monitoring workflows.

FastFacts - Real-Time AI Fact-Checking Desktop App (Hackathon Winner) October 2025
Electron, React, Flask, Opik, LangChain, Tavily, Gemini, OpenAI Speech-to-Text
- Developed a desktop application that performs real-time fact-checking on live system audio using an agentic AI workflow.
- Implemented an automated pipeline with OpenAI Speech-to-Text for transcription, Gemini for claim detection, and Tavily MCP for retrieving reputable online sources.
- Used Gemini to analyze evidence and classify claims as true, false, or unsubstantiated, returning a final verdict upon session completion.
- Integrated Comet's Opik to evaluate LLM outputs, ensuring consistency, reliability, and explainability across all agent interactions.